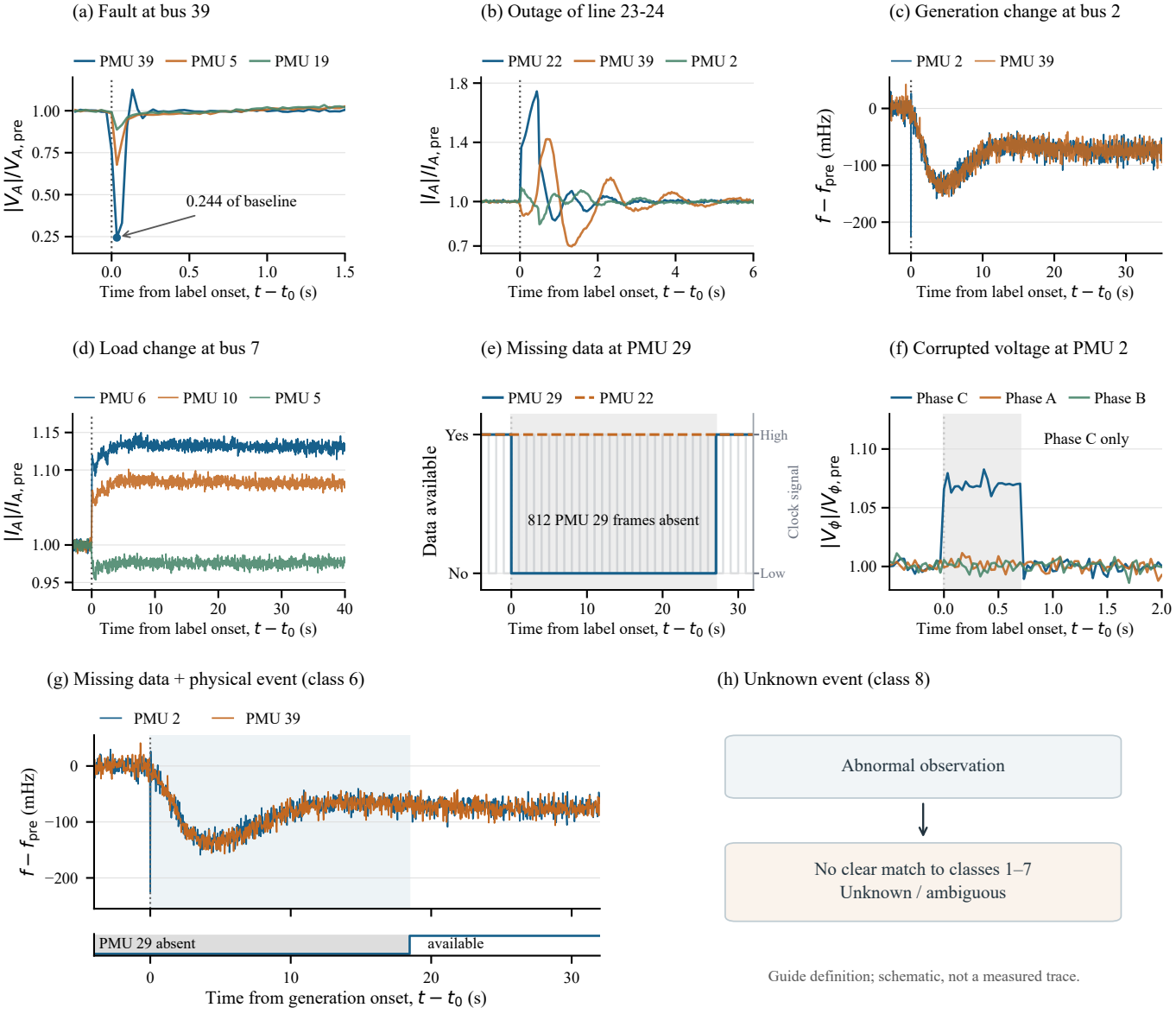

(a) Fault at bus 39
PMU 39
PMU 5
PMU 19
$|V_A|/V_{A,\mathrm{pre}}$
0.244 of baseline
Time from label onset, $t - t_0$ (s)
(b) Outage of line 23-24
PMU 22
PMU 39
PMU 2
$|I_A|/I_{A,\mathrm{pre}}$
Time from label onset, $t - t_0$ (s)
(c) Generation change at bus 2
PMU 2
PMU 39
$f - f_{\mathrm{pre}}$ (mHz)
Time from label onset, $t - t_0$ (s)
(d) Load change at bus 7
PMU 6
PMU 10
PMU 5
$|I_A|/I_{A,\mathrm{pre}}$
Time from label onset, $t - t_0$ (s)
(e) Missing data at PMU 29
PMU 29
PMU 22
Data available
Yes
No
812 PMU 29 frames absent
Clock signal
High
Low
Time from label onset, $t - t_0$ (s)
(f) Corrupted voltage at PMU 2
Phase C
Phase A
Phase B
Phase C only
$|V_\phi|/V_{\phi,\mathrm{pre}}$
Time from label onset, $t - t_0$ (s)
(g) Missing data + physical event (class 6)
PMU 2
PMU 39
$f - f_{\mathrm{pre}}$ (mHz)
PMU 29 absent
available
Time from generation onset, $t - t_0$ (s)
(h) Unknown event (class 8)
Abnormal observation
No clear match to classes 1–7
Unknown / ambiguous
Guide definition; schematic, not a measured trace.